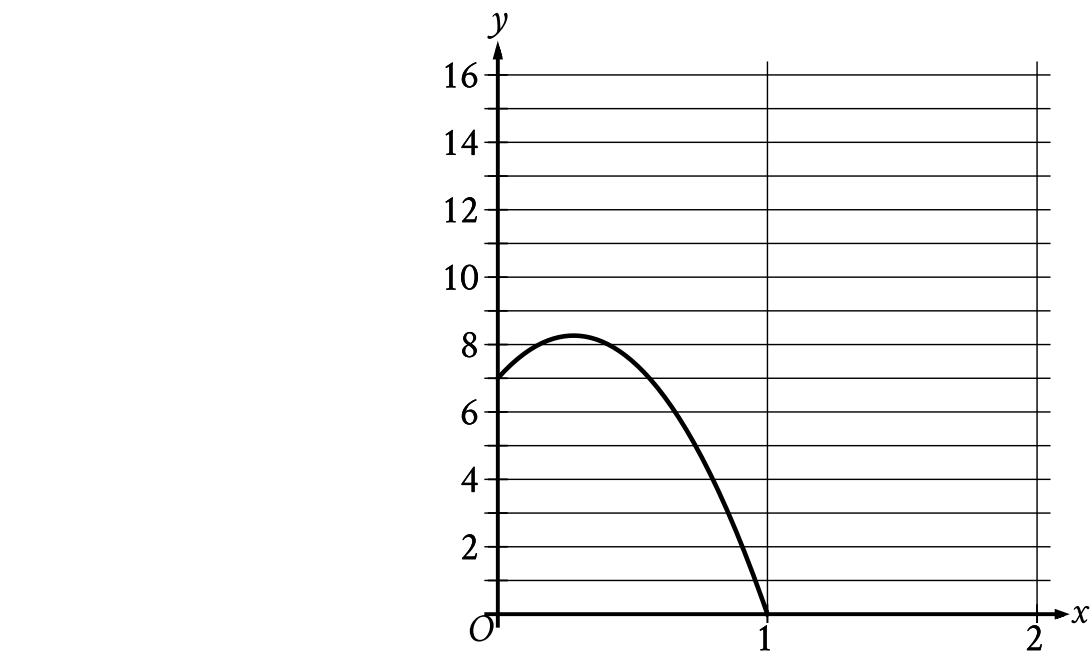


Question ID: 7fe7cf26

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

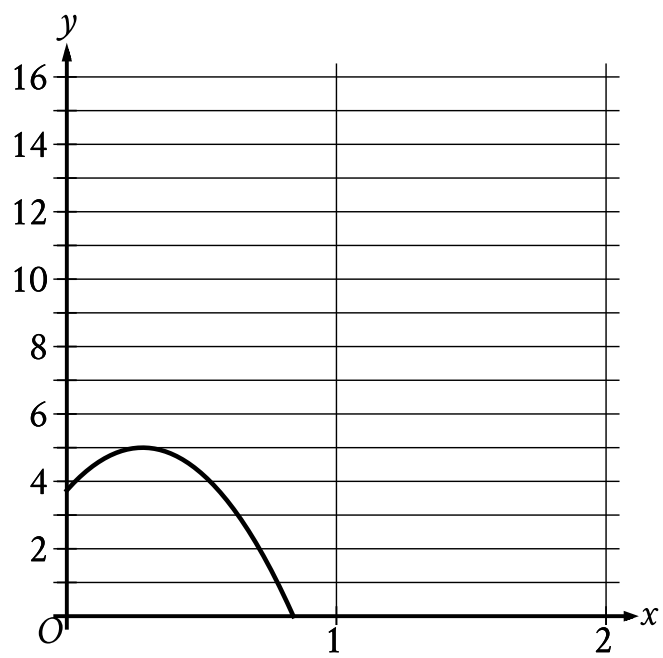
Question

During the first part of an experiment, a ball was launched from a **7-foot-tall** platform. The graph shows the height **y** , in feet, of the ball **x** seconds after it was launched during the first part of the experiment.

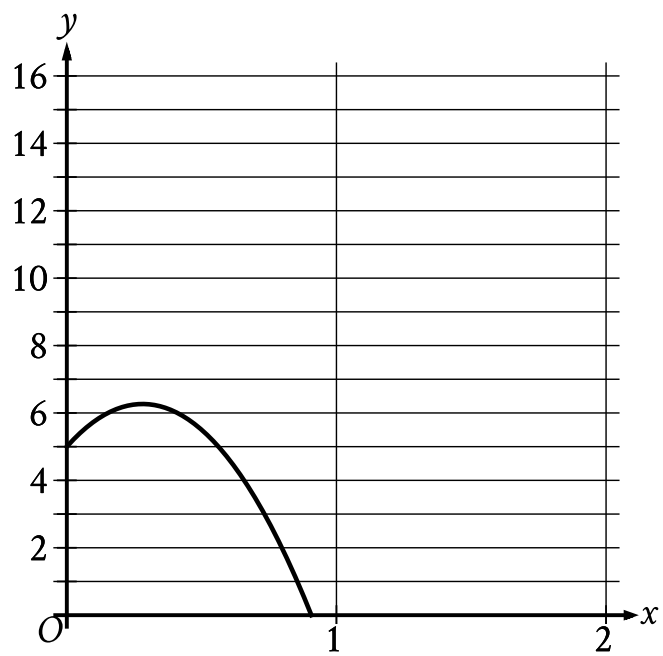


During the second part of the experiment, the ball was launched the same way, but from a platform that is **2** feet shorter than the first platform. Which of the following graphs could represent the height **y** , in feet, of the ball **x** seconds after it was launched during the second part of the experiment?

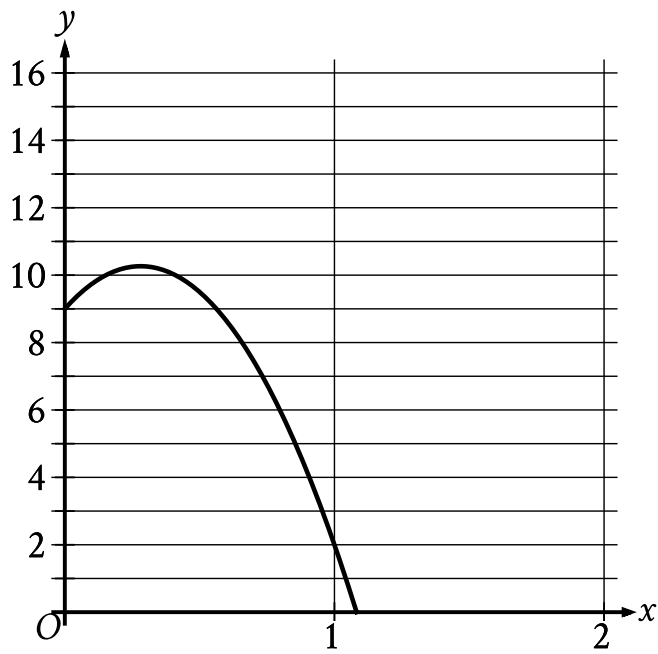
Answer



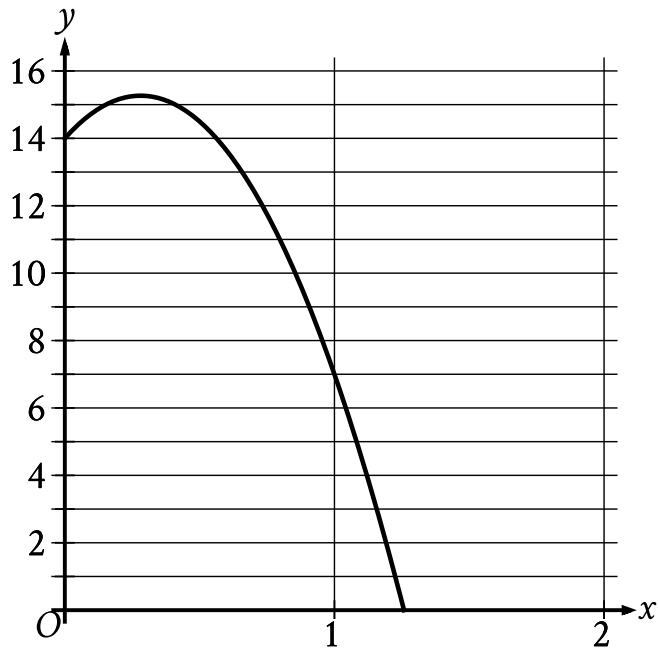
A.



B.



C.



D.

Correct Answer: B

Rationale

Choice B is correct. It's given that y represents the height, in feet, of the ball x seconds after it was launched. It's also given that during the first part of an experiment, a ball was launched from a **7-foot-tall** platform. Therefore, the y -coordinate of the y -intercept of the given graph, **7**, represents the platform height, in feet. During the second part of the experiment, the platform the ball was launched from was **2 feet** shorter than the platform in the first part of the experiment. It follows that the height of the platform in the second part of the experiment was **7 — 2 feet**, or **5** feet. Therefore, the y -coordinate of the y -intercept of the graph representing the second part of the experiment must be **5**. Only choice B satisfies this condition.

Choice A is incorrect. This could represent the graph if the ball were launched from a platform that was about **3** feet shorter rather than **2** feet shorter.

Choice C is incorrect. This could represent the graph if the ball were launched from a platform that was **2** feet taller rather than **2** feet shorter.

Choice D is incorrect. This could represent the graph if the ball were launched from a platform that was twice as tall rather than **2** feet shorter.